

$$u(x, y) = \sin(\pi x) \sin(\pi y),$$

$$u_{xx} = -\pi^2 u, \quad u_{yy} = -\pi^2 u,$$

$$-\Delta u = 2\pi^2 u = f, \quad u|_{\partial\Omega} = 0,$$

$$u\left(\frac{1}{2}, \frac{1}{2}\right) = 1, \quad \max_{\bar{\Omega}} u = 1,$$

$$\int_{\Omega} |\nabla u|^2 \, d\Omega = \int_{\Omega} f u \, d\Omega = \frac{\pi^2}{2}.$$